Research Questions

RQ1: *Does incorporating network topology and physics-based constraints improve fault detection and classification accuracy in district cooling networks, compared to topology-agnostic baseline methods?*

RQ2: *How does fault-diagnosis performance change when two faults occur concurrently at different locations, compared to the single-fault case?*

RQ3: *What architecture and interface would be required to integrate a topology-aware fault-diagnosis model into a digital-twin-as-a-service framework, and what practical constraints (latency, retraining, data flow) do that impose?*